



Mixed Gaussian-impulse noise reduction from images using convolutional neural network

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ABSTRACT

The removal of mixed-noise is an ill-posed problem due to high level of non-linearity in the distribution of noise. Most commonly encountered mixed-noise is the combination of additive white Gaussian noise (AWGN) and impulse noise (IN) that have contrasting characteristics. A number of methods from the cascade of IN and AWGN reduction to the state-of-the-art sparse representation have been reported to reduce this common form of mixed-noise. In this paper, a new learning-based algorithm using the convolutional neural network (CNN) model is proposed to reduce the mixed Gaussian-impulse noise from images. The proposed CNN model adopts computationally efficient transfer learning approach to obtain an end-to-end map from noisy image to noise-free image. The model has a small structure yet it is capable of providing performance superior to that of the well established methods. Experimental results on different settings of mixed-noise show that the proposed CNN-based denoising method performs significantly better than the sparse representation and patch-based methods do both in terms of accuracy and robustness. Moreover, due to the lightweight structure, the denoising operation of the proposed CNN-based method is computationally faster than that of the previously reported methods.

1. Introduction

Image denoising is a fundamental problem in image processing and computer vision. Images are corrupted during image acquisition or transmission due to inherent characteristics of imaging devices and transmission paths as well as due to defective equipment [1]. The goal of image denoising is to estimate the original noise-free image from its noisy observation. Image denoising as well as closely related operations like image inpainting [2] and watermark removal [3] are also recognized as preprocessing tasks for branches of computer vision such as image segmentation and pattern recognition. Two commonly encountered noise types in the literature are the additive white Gaussian noise (AWGN) and impulse noise (IN). The AWGN, which primarily originates from the sensor temperature and illumination levels of the environment, affects the entire set of pixels of an image [4]. On the other hand, the IN caused by faulty sensors or transmission errors, replaces certain image pixels with random values. The reasons for the generation of these noises are common and they often occur simultaneously resulting in mixed form of AWGN and IN. For example, AWGN and IN occur simultaneously in digital photography due to sensor temperature and faulty sensor triggering [5]. The noise in imaging of computed

tomography is often modeled as mixed AWGN-IN (see, for example, [6]). The complex mixing of photon and electronic noise, laser light reflection and dust on the glass slides are common in cDNA microarray imaging. In such a case, the noise in microarray images can be efficiently modeled as mixed AWGN-IN [7]. The denoising of images itself is regarded as an ill-posed inverse problem; and noticeably the estimation of a noise-free image becomes more challenging in the case of mixed-noise due to the fact that the distributions of the Gaussian and impulse noise differ significantly. Reduction of these noises have been studied extensively in the past decades separately as well as in the mixed form. In this section, the denoising methods for the AWGN, IN and mixed Gaussian-impulse noise are briefly reviewed. Finally, the scope in the area and the contributions of the work are given.

1.1. Related works

AWGN is widely studied noise model in the area of image denoising. Samples of a zero-mean Gaussian distribution are assumed to be added to pixels resulting in image corrupted with AWGN. A very common form of assumption for such a type of noise is the independent and

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identically distributed (i.i.d.) nature of the additive components. A good number of approaches have been investigated to restore images contaminated with AWGN. The noise can be reduced by using a simple approach of Gaussian filtering, in which case, however, the edge is not preserved giving rise to edge displacement, edge vanishing and even phantom edges [8]. Adaptive Gaussian filtering, bilateral filtering (BF), and non-local means (NLM) [9] consider this issue and show incremental accuracy in edge preservation. Since the inception of NLM, this approach has been rigorously investigated in a number of methods (see for example, [10,11]). Multiresolution analysis has also been used to reduce AWGN from images because of its success in 1D signal denoising [12,13]. For example, Rahman et al. [14] used the modified Gaussian-Hermite distribution to statistically model the discrete wavelet coefficients of images to reduce AWGN through Bayesian framework. Image denoising methods by using wavelets with improved directional selectivity such as the directionlet [15] and contourlet [16] were also tried. Dabov et al. [10] introduced block matching and 3D-filtering (BM3D) algorithm in which the similar image patches are grouped into a 3D matrix. Then, such groups are processed in a 3D transformed domain using one of the available sparse representations such as the DCT, bi-orthogonal and Haar wavelet coefficients. Liu et al. [11] prescribed a method using an adaptive soft-thresholding to find similar patches in images along with the l_2 -norm-based estimation for noise-free sparse coefficients. It was recommended that the block matching of BM3D should be carried out along edge direction to improve the denoising performance [17]. Xu et al. [18] developed a fast method based on the nonlocally centralized sparse representation algorithm. The neural network (NN) has also been employed to remove AWGN from images (see for example, [19–21]). The reduction of AWGN using the convolutional neural network (CNN) was recommended in [19]. Burger et al. [20] have provided a comparative study between the approaches of multilayer perceptron (MLP) and BM3D. Wang and Morel [22] used the mean-shift of the noisy patches and then attempted an MLP trained on a fixed strength noise level to remove Gaussian noise of variable strengths. Recently, a residual learning strategy has been adopted in CNN to reduce AWGN from images corrupted with unknown noise level [23].

Impulse noise is considered to be the replacement of certain portion of total pixels of an image by a set of fixed-level intensities with a given probability. Widely encountered impulse noise are salt and pepper impulse noise (SPIN) and random valued impulse noise (RVIN). In the case of SPIN, the corrupted image pixels get set at extreme values of the dynamic range of the image pixels, whereas a corrupted pixel in RVIN may be any of the random values within the range. In order to restore the images corrupted by IN, a number of nonlinear techniques have been tried. A simple rank order filter (ROF), e.g., the median filter (MF) [4] can be applied to detect and eliminate IN, but the local structures of images are destroyed by employing such a simple filtering technique for heavy noise corruption. In order to mitigate such problems, various improvements in MF such as the weighted MF [24], and center weighted MF (CWMF) [25] have been proposed. In general, these filters operate only on the corrupted pixels of an image, thus requiring a method to detect the noisy and noise-free pixels. The denoising methods using such an approach include the switching MF [26], adaptive MF (AMF) [27], and adaptive CWMF (ACWMF) [28]. The MLP scheme has also been used to restore images corrupted by IN (see, for example, [20]).

Tackling the mixture of AWGN and IN is relatively difficult because of the unique nature of each of the two types of noise. In general, the additive filters are successful in reducing the Gaussian noise, while the order statistics filters for the impulse noise [29]. In order to reduce the mixed Gaussian-impulse noise from images, conventional methods use rank order statistics of pixels to detect and remove IN first, and then use a separate denoising technique by assuming that the residual noise is Gaussian. The order statistics filters when applied to images corrupted with mixed AWGN-IN produce grainy and visually unpleasant results, and successive Gaussian filters cannot compensate such noise effectively [30]. The alpha-trim mean filters provide both

the additive and rank order properties. However, under heavy noise, these methods provide inadequate performance as the local structures of images become smeared. In order to preserve the details in an image, the BF has been extended as a trilateral filter to reduce the mixed-noise by incorporating the rank-ordered statistics of the absolute differences of intensities of neighboring pixels [30]. Cai et al. [31,32] proposed a two-phase method to estimate the noise-free images. In this method, IN is detected and removed first by using AMF, and then the resultant image is denoised by optimizing an l_1 -norm-based regularization function. Cai's method can be treated as a rank order filtering technique, provided the regularization parameter is very small. This method has been studied for mixed-noise removal by exploring different kinds of norms and regularization techniques in the second phase. The norms that have been investigated in the optimization technique include the total variation norm of wavelet coefficients of images [33]. Liu et al. [34] used the dictionary learning model with sparse representation to estimate the noise-free image. Agostinelli et al. [21] employed the ensemble of stacked sparse autoencoders and adaptive averaging technique for image denoising. Zhang et al. [35] proposed an iterative split-Bregman-based denoising algorithm that requires joint statistical modeling of similar patches in an image. In a unified framework of joint detection of noisy pixels and reduction of noise components, Xiang et al. [36] proposed an iterative dictionary learning-based method called the weighted encoding with sparse nonlocal regularization (WESNR). The method provides a good performance in preserving the local structures for relatively smooth images, but falls short when images have higher details. Dictionary learning method has recently been used for hyperspectral image super resolution under mixed Poisson–Gaussian noise [37].

1.2. Scope of analysis

To the best of our knowledge, the removal of mixed-noise from images has not yet been attempted by using CNN, which has shown immense success not only in image classification [38] but also in different regression-type problems such as image enhancement [39], super-resolution [40], blind deconvolution [41], and inpainting [42]. The increasing interest of CNN is also largely due to the availability of application-oriented large databases and efficient parallel computing in graphics processing units (GPUs) [38]. Convolutional neural network has been attempted to remove AWGN from images (see [19], and [23]). Two stream NNs have also been employed for reducing AWGN and IN independently by training the patches of certain test images [21]. The success of CNN in many image processing techniques as well as a few instances of application of NNs in simple problems of image denoising has thus motivated us to develop a single-stream CNN architecture in order to tackle the highly challenging problem of removal of mixed-noise by generating the reconstruction filters with the consideration of large-scale image variabilities. A question may arise as to why CNN should be chosen as compared to the acclaimed denoising methods such as those that adopt sparse representation and patch based denoising. To answer this question, first we would like to refer to the methods adopting the sparse representation. Recently, it has been shown that CNN possesses higher representation capability as compared to traditional sparse representation, thus provides better performance in image super-resolution [40]. Traditionally, the sparse dictionaries are constructed by vectorizing the image matrices or patches, and thus 2D structural information, i.e., dependency of pixels of local neighboring regions may be lost in dictionary learning. On the contrary, CNN is capable of maintaining the 2D structural information both in the training and testing phases, since the convolution operation considers the local neighboring image pixels by using 2D masks. Second, the patch-based methods such as NLM and BM3D use self-similarity regularization and thus require a computationally heavy iterative optimization algorithm. In addition, the performance of these methods can be sub-optimal, if the images have a low number of self-similar patches. On the contrary, CNN optimizes its weights of the convolution masks through gradient-based training scheme, which inherently considers self-similarity in the

entire set of patches available in relatively large number of training images. In the training phase of a CNN-based network, the weights of the masks are learned through the gradients of local neighboring pixels in such a way that the noise is reduced. Since a huge number of samples as well as large variabilities in images are used during training, the CNN-based denoising is expected to be optimal. Moreover, the performance of CNN has been shown to have significantly improved due to the inclusion of newer activation functions like the rectified linear unit (ReLU) [38] and newer learning algorithms such as the root mean square propagation (RMSProp) [43]. Such a success has not been observed in other types of neural networks. For example, the long short term memory network [44], which is primarily used to model time series data, has not benefited much due to ReLU activation function, as CNN has. In addition, regression type tasks including denoising require images to be processed in patches prior to inputting them to the network. In such a case, the implementation of CNN using modern GPUs to train an end-to-end model is more efficient than that of MLP. Further, once a CNN is trained, the weights learned by the network can be transferred to a closely related network that has a similar setting for faster learning [45]. Thus, the removal of mixed-noise using CNN is a viable idea and the development of efficient method of noise reduction using a suitable network architecture is worth investigating.

1.3. Specific contributions

The main objective of this paper is to present a CNN architecture for reduction of mixed-noise encountered in practice. We show that a moderate size CNN is capable of efficiently removing mixed Gaussian-impulse noise from images. Overall, the contributions of this paper are as follows:

- A four-stage CNN architecture is proposed for reduction of mixed Gaussian-impulse noise in images. The network learns end-to-end mapping from noisy images to noise-free estimates with trivial pre-processing.
- Faster training of network is obtained by adopting the mechanism of transfer learning.
- The performance of the proposed CNN-based method is evaluated using sufficiently large datasets as well as commonly-referred test images. The overall denoising performance in terms of accuracy and robustness is shown to be better than that of the state-of-the-art methods considering different settings of mixed-noise.

The paper is organized as follows. Section 2 provides the problem formulation. Section 3 presents the proposed CNN-based method for reducing the mixed-noise. This section provides a brief review of CNN layers, and the proposed architecture and training scheme. Section 4 gives the details of the experiments and the results obtained. Finally, Section 5 provides some conclusions.

2. Problem formulation

Let $\mathbf{X}$ be a noise-free image of size $M_v \times M_h$ with each element being represented as $x(m_v, m_h)$ at pixel location (m_v, m_h) , where $m_v \in \{1, 2, \dots, M_v\}$ and $m_h \in \{1, 2, \dots, M_h\}$. Let $\mathbf{X}_n$ be the noisy observation of the image $\mathbf{X}$ with a relation given by

$$\mathbf{X}_n = f(\mathbf{X}) \quad (1)$$

where $f(\cdot)$ is the degradation function. Also, let a noisy pixel be denoted as $x_n(m_v, m_h)$. We consider two types of corruption: (1) mixed AWGN and SPIN and (2) mixed AWGN, SPIN and RVIN. If the image is corrupted only by AWGN, then a noisy pixel is given by

$$x_n(m_v, m_h) = x(m_v, m_h) + v(m_v, m_h) \quad (2)$$

where $v(m_v, m_h)$ is a sample of i.i.d. zero-mean Gaussian distribution with standard deviation σ . Within the given dynamic range, let the maximum and minimum values of an image pixel be d_{max} and d_{min} , respectively. Then, an image is corrupted with SPIN when $x_n(m_v, m_h)$ is either d_{max} or d_{min} with equal probability $p/2$ ($p \leq 1$). Thus, for mixed AWGN + SPIN noise a pixel is corrupted by AWGN with a probability $(1 - p)$. Using these definitions, the noisy observation for each pixel of mixed AWGN + SPIN can be described as [36]

$$x_n(m_v, m_h) = \begin{cases} d_{min} & \text{with probability } p/2 \\ d_{max} & \text{with probability } p/2 \\ x(m_v, m_h) + v(m_v, m_h) & \text{with probability } 1 - p \end{cases} \quad (3)$$

In a similar fashion, let an image be corrupted with RVIN when $x_n(m_v, m_h)$ attains a random value $d(m_v, m_h)$ with probability r ($r \leq 1$). The value $d(m_v, m_h)$ is uniformly distributed within the dynamic range $[d_{min}, d_{max}]$. Using the definition of RVIN, the noisy observation for each pixel of mixed AWGN + SPIN + RVIN can be described as [36]

$$x_n(m_v, m_h) = \begin{cases} d_{min} & \text{with probability } p/2 \\ d_{max} & \text{with probability } p/2 \\ d(m_v, m_h) & \text{with probability } r(1 - p) \\ x(m_v, m_h) + v(m_v, m_h) & \text{with probability } (1 - p)(1 - r) \end{cases} \quad (4)$$

Fig. 1 shows a typical image and its noisy version corrupted with AWGN and SPIN, and the corresponding histograms of the images. The noisy image is obtained by using the AWGN parameter $\sigma = 10$ and SPIN parameter $p = 0.3$. It is observed from this figure that the degradation function changes the original histogram significantly. In particular, the variation of histogram in the dynamic region has been smoothed significantly primarily due to AWGN. At the same time, two strong peaks have appeared at the two ends of the histogram due to the presence of SPIN. In other words, the effects of mixed-noise on the histogram of image are contrasting in nature. The goal of denoising is to find an estimate $\hat{\mathbf{X}}$ for the noise-free image $\mathbf{X}$ from its noisy observation $\mathbf{X}_n$.

3. Proposed CNN-based denoising

In this paper, a feed-forward CNN is employed in order to remove mixed-noise from an image. A typical CNN can have different types of layers in its architecture. Let L be the number of layers in a network. Let the output of a layer i ($i \in \{1, 2, \dots, L\}$) be denoted as $\mathbf{X}_i$ with dimension $C_i \times M_{vi} \times M_{hi}$, where C_i is the depth parameter that is often referred to as the number of channels in the output. The vertical and horizontal dimensions, namely, M_{vi} and M_{hi} , are simultaneously called spatial parameters of the output. In the feed-forward architecture, the input of the i th layer is taken from the output of the $(i - 1)$ th layer. For the first layer, the input is the noisy image, i.e., $\mathbf{X}_0 = \mathbf{X}_n$, and for the last layer, the output is the estimate $\hat{\mathbf{X}}$ of the noise-free image. In this section, first we briefly describe the typical layers of CNN that are used for different types of classification or regression problems. Next, we present the proposed CNN-based denoising method that uses the combination of these layers.

3.1. Layers of CNN: a brief review

The typical layers of a CNN are the convolution layer performing the filtering operation, the ReLU layer performing the activation, and the max-pool layer reducing the spatial dimensions. The layers are described sequentially in brief as follows:

3.1.1. Convolution layer

This layer performs the 2D convolution operation on the input data $\mathbf{X}_{i-1}$ using a set of layer-dependent filters. Let $\mathbf{W}_i$ be a filter set with dimension $C_i \times C_{i-1} \times N_i \times N_i$, where C_i and C_{i-1} are the number of channels of the output and input of this layer, respectively, and N_i

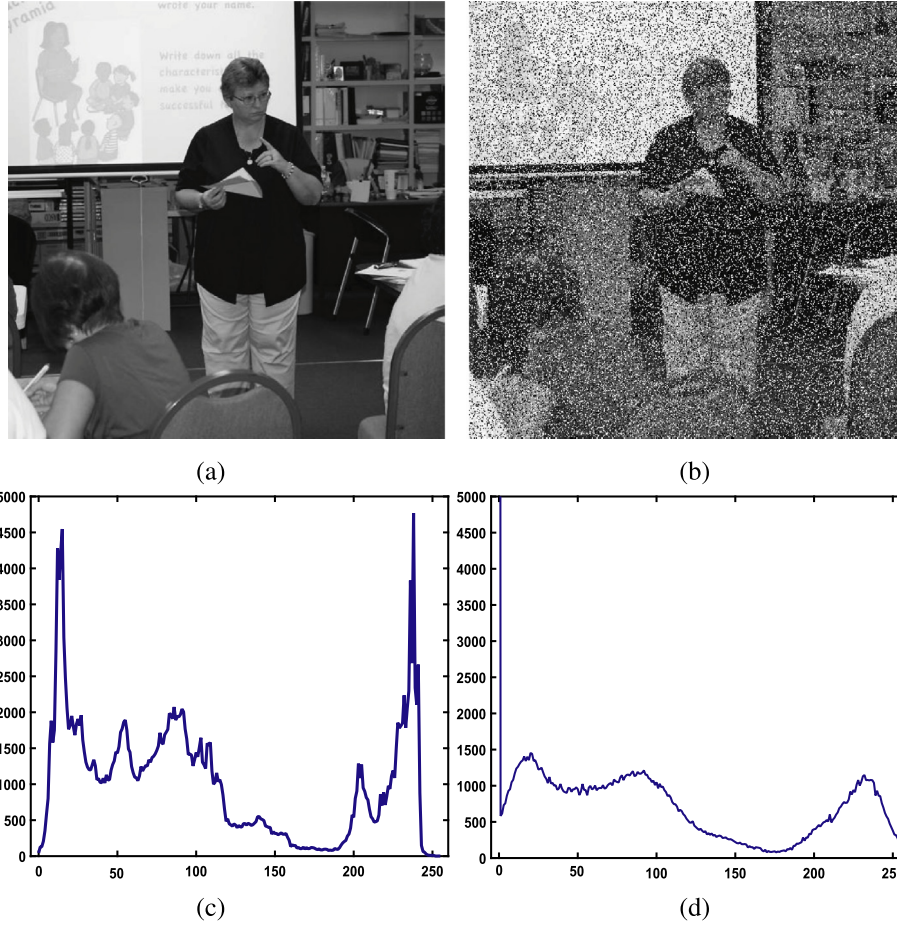


Fig. 1. Effect of mixed AWGN-SPIN noise on a typical image and its histogram. (a) Noise-free image, (b) noisy image, (c) histogram of noise-free image, and (d) histogram of noisy image.

is the square-size support parameter of the filters. The parameter C_i represents the number of filters in the set $\mathbf{W}_i$. Each of the filters has a corresponding bias term, resulting in a bias vector $\mathbf{b}_i$ with C_i number of elements. Hence, the output of this layer is obtained from the output of the previous layer, the bias term, and the corresponding filter set as

$$\mathbf{X}_i = \mathbf{W}_i * \mathbf{X}_{i-1} + \mathbf{b}_i \quad (5)$$

where $*$ represents the linear convolution operation. This operation results in the dimension of output $\mathbf{X}_i$ to be $C_i \times M_{vi} \times M_{hi}$ from input $\mathbf{X}_{i-1}$ with shape $C_{i-1} \times M_{v(i-1)} \times M_{h(i-1)}$. There is a parameter called ‘stride’, which can be used in the convolution layer to increase or decrease the spatial dimensions of the output. The spatial dimensions remain the same from the input to the output, when the parameter is set to unity. A value of the stride parameter greater than unity decreases the dimensions of the output, whereas a value less than unity increases the dimensions. In the case of dimensionality reduction in the output, however, a general tendency is to set the stride parameter to unity and to use the pooling layer to perform the downsampling operation. In the graphical stick diagram, the convolution layer is shown by a rectangle.

3.1.2. Rectified linear unit layer

It is referred to as the activation layer, wherein only the elements of input with non-negative values are transmitted to the output and the rest are set to zero. In other words, the input–output relation of this layer is given by

$$\mathbf{X}_i = \max(\mathbf{0}, \mathbf{X}_{i-1}) \quad (6)$$

where $\mathbf{0}$ is a zero-matrix with same size as that of $\mathbf{X}_i$, by considering the fact that the dimensions of the input and output of this layer remain

the same. It is well known that the rectification aids the convergence of the neural network much faster as compared to the traditional sigmoid function does. In addition, the ReLU unit assists the neural network to attain a better sparse representation (see [46]). It is customary that the convolution layer be followed by the ReLU activation. In order to represent this layer, a solid line is used in the graphical stick diagram.

3.1.3. Max-pool layer

In this layer, a non-linear downsampling operation is carried out between the input and the output. A mask of size $K_i \times K_i$ is selected along the spatial dimensions of the input in a non-overlapping manner, and then the maximum value of the data in the mask is passed on to the output layer. The pooling layers are reported to increase the robustness of CNN-based algorithms in the presence of noise and clutter, and to be particularly well suited for attaining sparsity in the data [47]. In practice, the max-pool layer is preceded by a ReLU layer. To describe this layer, the notation MP_{K_i} is used in the model description and a converging trapezoid in the stick diagram.

3.2. Proposed model

In general, a CNN-based method produces an end-to-end model with trivial pre- and post-processings, which are usually referred to as the non-trainable operations. In many cases, an application-oriented pre-processing can boost the performance of the neural network significantly (see for example, the usage of bicubic interpolation applied for super-resolution in [40]). In the literature, almost all the methods use an ROF such as MF as a preprocessor to denoise images corrupted by mixed Gaussian-impulse noise. We have also adopted such a practice, and

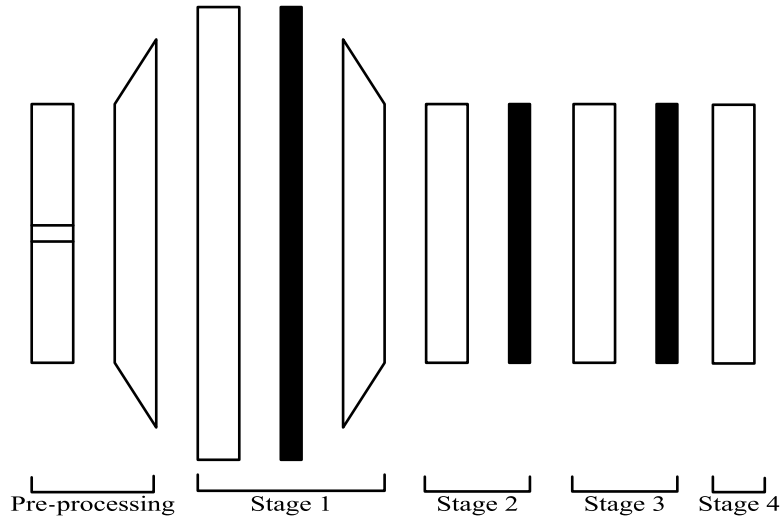


Fig. 2. Stick diagram of the proposed feed-forward CNN architecture showing the preprocessing steps and 4-stage convolution filtering. Operations from the left to right is considered to be the forward path. Each of the processing steps or layers of the network is represented by its corresponding geometric shape. The rank order filtering operation is shown by a stripe in a rectangle, the upsampling operation through bicubic interpolation by a diverging trapezoid, the convolution layer by a rectangle, the ReLU layer by a solid line, and the max-pooling layer by a converging trapezoid.

hence, the first step of the proposed model to denoise the corrupted image X_0 is given by

$$X_1 = ROF(X_0) \quad (7)$$

where ROF denotes a suitable rank order filter or combination of such filters. The ROF can be formed by using existing filters such as MF, AMF, CWMF, ACWMF or Cai's method [31] that are successful for reducing IN. The choice of such a filter largely depends on the type of IN corrupting the image. In [36], it is suggested that ROF be AMF when the noise is SPIN, and MF when the noise is SPIN + RVIN. In the proposed model, the Cai's method [31] has been chosen as ROF when the impulse noise is SPIN, and AMF followed by ACWMF has been chosen when the noise is SPIN + RVIN. In the stick diagram, we recommend the use of a stripe in a rectangle to represent ROF.

To feed the convolution layer with a slightly smoother version of the noisy image, an upsampling operation is performed on the rank order filtered image using the bicubic interpolation (BI). In other words, the input-output relation of the second layer of the proposed model is given by

$$X_2 = BI(X_1) \quad (8)$$

where BI denotes the interpolation function. It can be shown that the frequency response of the interpolation functions exhibit a nature of a low-pass filter [48]. Thus, some high frequency components that arise from the rank order filtering on the Gaussian noise are mitigated using such functions. In the stick diagram, we prefer to use a diverging trapezoid to represent this interpolation function.

In order to denoise the rank order filtered and interpolated image, a 4-stage convolution filtering scheme is employed in the proposed model. A natural question that may arise is as to why 4 stages of convolutional layer are chosen in the model. In this context, an ablation study using 1000 images is performed by varying the number of convolutional layers from 2 to 6 in the proposed architecture. It is observed that the relative improvements of denoising performance in terms of mean of the peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [49] decrease with the number of stages. Fig. 3 shows the relative improvements of PSNR and SSIM in percentage as well as the standard deviation of these metrics as the number of stages increases. It is seen from this figure that the increase in PSNR with the addition of a new convolution layer is not so significant when the number of layers is more than 4. In particular, the increase in PSNR, i.e., $\Delta PSNR$, due

to the inclusion of a new convolutional layer over 4 is less than 0.5%. Similar observation is made for the increase in SSIM, i.e., $\Delta SSIM$, due to the addition of a new convolution layer. Further, it is observed that increasing the number of convolutional layers not only increases the computational load significantly, but also decreases the robustness in terms of increasing the standard deviation of both the metrics. Thus, it is concluded that a CNN architecture with 4 convolution layers is adequate to provide a satisfactory level of denoising performance both in terms of accuracy and robustness. In each of the four stages, the convolution layers may be followed by a ReLU or max-pool layer. Since the spatial dimensions of a noisy image are increased due to interpolation in the preprocessing step, at least one downsampling operation using the max-pool layer is required in the overall model. In this context, only the first stage convolution filtering uses all three kinds of layers. In particular, the output of the convolution layer is fed to a ReLU activation layer. This output is then max-pooled and spatial dimension is reduced. Thus, the output of the first stage CNN filtering can be written as

$$X_3 = MP_{K_1}(\max(0, W_1 * X_2 + b_1)). \quad (9)$$

It is evident that the spatial dimensions of the output of the first stage convolution filtering are the same as that of the noisy input image. The second stage of the proposed model uses only the convolution and ReLU layers providing the output given by

$$X_4 = \max(0, W_2 * X_3 + b_2). \quad (10)$$

The third stage of the network is a repetition of the second, and thus, the output is given by

$$X_5 = \max(0, W_3 * X_4 + b_3). \quad (11)$$

In the final stage, only the convolution layer is used in the model. This stage provides the estimate of the noise-free image having the same dimensions as that of input as

$$\hat{X} = W_4 * X_5 + b_4. \quad (12)$$

Fig. 2 shows the stick diagram of the proposed feed-forward CNN-based image denoising method with distinct marking of the preprocessing steps and 4-stage convolutional filtering. The geometric shapes of the preprocessing steps have been specified in the beginning of this section, whereas that of the CNN layers follow the convention. It is seen from this figure that the architecture of the proposed CNN model is relatively small, and thus, it is implementable even in a light weight computational system.

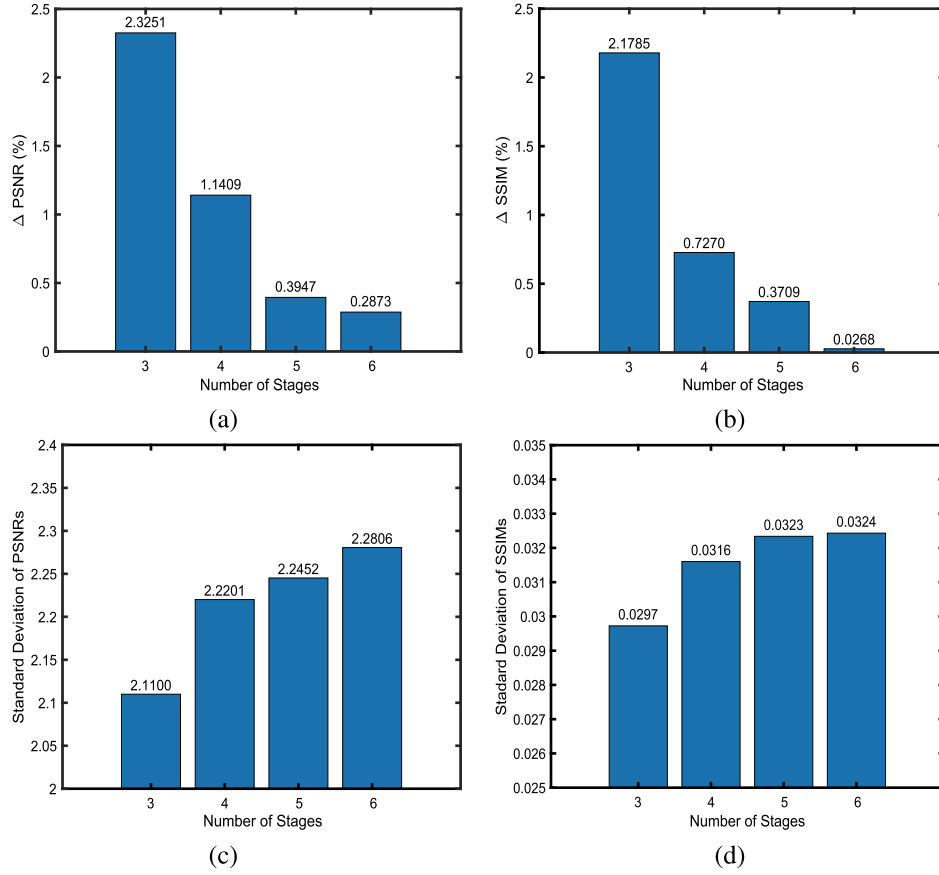


Fig. 3. Denoising performance by varying number of stages in the CNN architecture. (a) Increase of PSNR from previous stage. (b) Increase of SSIM from previous stage. (c) Standard deviation of PSNRs. (d) Standard deviation of SSIMs.

3.3. Training scheme

In order to obtain the end-to-end mapping function to denoise the images, the parameters of the proposed CNN model are required to be evaluated. The filters W_1 , W_2 , W_3 and W_4 and bias terms b_1 , b_2 , b_3 and b_4 in the four convolution layers of the proposed model are the parameters that have to be determined through learning. In order to train the network with the back-propagation algorithm, a differentiable and tractable loss function is required. The choices of loss function for such a regression task include the l_1 norm, l_2 norm, and Frobenius norm. The Frobenius norm is defined for matrices, and thus suits well when working with images. Moreover, the square of the norm is preferable due to numerical stability of the differentiation. Thus, we choose the squared Frobenius norm as the loss function in order to learn the network, and therefore, to estimate the parameters of the mapping function.

In general, in order to learn the CNN model, different types of data augmentation techniques are adopted during the training session [38]. This is mainly due to the fact that an optimal set of parameters of the network are achieved by increasing the number of image variabilities seen by the network. The augmented set of noisy images are fed to CNN and the corresponding noise-free images are estimated. Then, the loss function is defined in terms of an original image X and the corresponding estimate $\hat{X}$ as

$$D = \|X - \hat{X}\|_F^2 \quad (13)$$

where $\|\cdot\|_F$ is the Frobenius norm. Elements of the convolution filters, denoted by w , are updated using the gradient of the loss function D with respect to w . The update is performed in terms of the weighted average of the square of the gradient, which is also referred to as the

mean squared given by [43]

$$MS(w(\eta)) = \gamma MS(w(\eta-1)) + (1-\gamma) \left(\frac{\partial D(\eta)}{\partial w(\eta)} \right)^2 \quad (14)$$

$$w(\eta+1) = w(\eta) - \frac{\lambda}{\sqrt{MS(w(\eta)) + \epsilon}} \frac{\partial D(\eta)}{\partial w(\eta)} \quad (15)$$

where λ ($\lambda > 0$) and γ ($0 < \gamma < 1$) are hyper-parameters known as the learning and decay rates, respectively, η is the iteration number, and ϵ is a numerical stability factor. In a similar fashion, the elements of the bias terms are also updated in the training phase. It is noted that once a CNN is trained for a certain set of noise parameters, the filters and bias terms learned by the network can be used to initialize the same with new set of noise parameters, and thus achieving faster learning of the networks [45].

4. Experiments

Experiments are carried out to evaluate the performance of the proposed CNN-based method by comparing it with that of the existing ones in reducing the mixed Gaussian-impulse noise from images. In this section, first we describe the datasets used in the experiments. Next, the data augmentation processes, which are required to achieve generalized results, are described. The setup of the proposed CNN model and the initialization of the parameters are presented in the subsection that follows. The representations of filters learned in the training phase are given in the following subsection. Then, the denoising methods that are compared with the proposed CNN-based method are described briefly. Finally, the comparative results on the datasets in terms of the performance metrics, visual quality, and computational load are presented.

4.1. Datasets

The construction of training and testing sets is very important for learning-based methods. A natural question arises as how to define these sets. In this context, first we would like to emphasize that the reporting of average denoising performance on a large set of test images is preferable over that of ordinary performance on a small number of recurrently appearing images in the literature. In other words, the testing sets should be relatively large so that one can have more realistic evaluations of the performance. Second, the deep learning produces generalized results when the training dataset is comprehensive. Thus, we ensure that the training dataset is sufficiently large. Finally, for a learning-based method, the performance should be more authentic, if the training and testing datasets are mutually exclusive and at the same time they are collected from different sources. In order to meet these requirements, we ensure that both the training and testing datasets in the experiments are sufficiently large and they are constructed from two different databases. In particular, we consider the Places2 [50] and ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2015 object detection (DET) [51] databases, which are well known in the area of computer vision and image processing. The first database is chosen for training and validation, and the second one for the testing purpose. The experiments are conducted by converting the color images to grayscale images.

The training dataset is constructed from the Places2 database by randomly choosing 2×10^4 images. It has been reported that a CNN model can reasonably learn from a smaller sample size (~ 100) for a regression-type problem as compared to a larger sample size ($\sim 10^5$) for a classification-type problem (see, for example, [40]). Hence, the size of training set as considered in the experiment of image denoising is reasonable. In addition, to obtain generalized results, we adopt data augmentation techniques (see details in Section 4.2), which eventually increases the number of images several times in the training phase. In order to track the improvement of denoising after each epoch of the training phase, 50 mutually exclusive images are selected from the Places2 database as a validation set. Further, to facilitate the computation of the batch gradient descent technique used in CNN, it is required that the size of the images of the training and validation sets is the same, and preferably their sizes are small. In this context, each image from the training and validation sets is changed to a square shape by cropping it from the center while keeping the smaller dimension of the original image intact, and then the images are resized to 128×128 . In the experiments on mixed AWGN–SPIN, the noisy versions of the images are generated by contaminating the processed images following (3). We have considered four values of the AWGN parameter, namely, $\sigma = 10, 15, 20$, and 25 , and two values of the SPIN parameter, namely, $p = 0.15$ and 0.30 that result in eight different training and testing scenarios. For experiments involving the mixed AWGN + SPIN + RVIN, the noisy images are generated by using (4), wherein the noise parameters are chosen as $\sigma = 10, p = 0.20$ and $r = 0.05$, and $\sigma = 20, p = 0.15$ and $r = 0.10$.

In order to test the performance of the trained network, generic test datasets are constructed by choosing 1000 images from the ILSVRC 2015 DET database. The chosen images have good mixtures of smooth regions, edges, and textures. The test sets are obtained by considering three practical scenarios described below:

1. Test Set 1: Similar to the training set, 1000 images are converted to grayscale images of size 128×128 . The noisy test images are generated with noise parameters that are the same as for the training phase.
2. Test Set 2: The same images as in Test Set 1 are selected in this scenario, but the attributes of the noise parameters are chosen differently. This set analyzes the performance of the proposed method when the noise parameters in the testing phase deviate from that in the training phase. In particular, the parameters σ, p

and r are varied up to $\pm 10\%$ from their nominal values randomly for each of the images. The values for σ, p , and r are drawn from uniform distributions of $\mathcal{U}(0.9\sigma, 1.1\sigma)$, $\mathcal{U}(0.9p, 1.1p)$, and $\mathcal{U}(0.9r, 1.1r)$, respectively, for each image.

3. Test Set 3: This test set is generated with a view to obtaining the performance of the network for arbitrary size images. In this scenario, 15 images are chosen from the testing set such that the dimension without resizing has the maximum variation with respect to the training set. In the experiments, the dimension of these images varies from 146 to 500 and 226 to 500 along the horizontal and vertical axes, respectively. The value of the noise parameters, namely, σ, p , and r , are kept the same as that in the training dataset.

In order to investigate the denoising performance of the proposed method on individual images, the commonly-referred test images are also considered in the experiments.

4.2. Data augmentation

Data augmentation results in the trained filters for denoising more robust to unseen images. In the experiments, a number of augmentation techniques are employed on the training images. The operations include the transpose, and the horizontal and vertical flips of the image matrices. The probability of each of the operations is set to 0.5. Thus, the combination of the operations produces up to 8 variants of each of the images. As a result, different orientations of the same image is available to the network for learning. Moreover, the samples of the noise sequence are altered after every 20 epochs in the training phase. Consequently, different instances of the noise for each of the training images are seen by the network. In effect, the size of the ultimate training dataset turns several times larger than that of initial set, which helps the network to attain generalization.

4.3. Model setup

The values of the parameters of the proposed CNN model described in Section 3.2 are chosen as per the dimensions of the input and output as well as those recommended in practice. Since the experimental images in the training and test datasets are grayscale in nature, the parameter C_0 in the input channel at the very beginning is set to 1. The proposed four-stage network has four filter sets, namely, $\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3$ and $\mathbf{W}_4$, for which the number of filters in each set and that of the weights in each filter are assigned as per the number of output channels and the kernel size used, respectively, in a layer. By the definition of a convolution layer, the number of output channels and that of the bias terms for the layer are kept equal. In the experiments, the channel parameters C_1, C_2 and C_3 , corresponding to the filter sets $\mathbf{W}_1, \mathbf{W}_2$ and $\mathbf{W}_3$, are set to 100, 200 and 100 when $\sigma < 20$ and to 100, 500 and 100 when $\sigma \geq 20$, respectively. The filter set $\mathbf{W}_4$ is in the last stage of the network, and the corresponding channel parameter C_4 specifies the output channel. This channel parameter is set to 1 as the output provides the estimated noise-free image of the corresponding noisy input. The kernel sizes of the filters are chosen based on the strength of Gaussian noise. It has been observed that a larger kernel size is beneficial in denoising for a higher value of σ . In addition, the kernel size N_1 of the filter set $\mathbf{W}_1$ is set to a higher value so that information can be extracted from a relatively larger neighboring region of an upsampled image in the first stage. The kernel size of the following three filter sets, namely, $\mathbf{W}_2, \mathbf{W}_3$ and $\mathbf{W}_4$, is set to a lower value as compared to N_1 in order to process a smaller local region of the filtered input, which has the spatial dimensions the same as that of the original image. Independent of the noise strength, the quadruplet (N_1, N_2, N_3, N_4) is chosen as (7, 5, 5, 3). The parameter K_1 of the max-pool layer is set to 2, which is a traditional value. This value of the parameter also results in the spatial dimensions of the final output to be the same as the original image from the upsampled version of the output of the first convolution and ReLU layers.

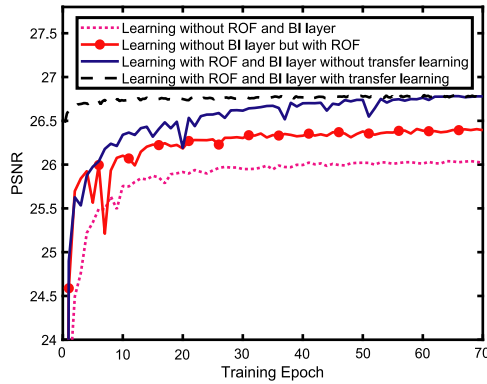


Fig. 4. Learning curves of the proposed CNN-based model trained for removal of mixed AWGN + SPIN with noise parameters $\sigma = 10$ and $p = 0.30$. The pink dotted learning curve is obtained when both the ROF and BI layers as well as the subsequent MP_{K_i} layer are removed from the model. The solid red curve with circle markers results in when the network uses the ROF layer without the BI layer. The blue solid curve is obtained when the network uses both the ROF and BI layers but without any prior information for initialization of weights and biases. The black dashed curve is obtained when both the ROF and BI layers are employed and at the same time transfer learning is adopted using the known weights and biases that are trained for $\sigma = 10$ and $p = 0.15$.

The initialization of the weights and biases of the network and that of the parameters of learning mechanism are important for efficient training. When no prior information about the weights W_1, W_2, W_3 and W_4 and the corresponding biases b_1, b_2, b_3 and b_4 is available, they are initialized by taking samples from a zero-mean Gaussian distribution with standard deviation 10^{-3} . In order to slow down the gradient descent and to avoid shoot off as the network converges, a decreasing value of the learning rate λ is chosen. Empirically the initial value of λ is set to 10^{-4} for first 10 epochs, and then its value is decreased by 5% after each epoch. The decay rate γ is set to a value of 0.9 as recommended in [43]. The numerical stability factor ϵ is set to a small value of 10^{-8} in order to avoid division by zero, and the remaining term MS is initialized to zero.

When a prior information on the weights and biases of a network is available, the network can be trained in a closely related setting using the transfer learning method [45]. In this paper, the prior information on weights and biases of the proposed network is that of the network trained on different noise settings. For example, if the weights of the network trained for $\sigma = 10$ and $p = 0.15$ are available, then the network for $\sigma = 10$ and $p = 0.30$ can be trained for this new set of noise parameters by fine-tuning the known weights. While the network takes a large number of epochs to achieve a considerable denoising performance without prior information, it would take only a few epochs to train the network if the initialization considers the known weights. It is also observed that the initial learning rate λ of the networks adopting transfer learning can be lower than that of the networks trained without prior information. In the experiments, the value of λ is set to 10^{-5} for transfer learning.

Fig. 4 shows the learning curves of the proposed model in terms of the PSNR obtained by using the validation set for 70 epochs in four different scenarios. The blue solid curve represents the learning curve for the four stage architecture trained for removal of mixed AWGN + SPIN with noise parameters $\sigma = 10$ and $p = 0.30$ without any prior information of the weights and biases. The solid red curve with circle markers depicts the learning curve when only the ROF layer is employed, but the upsampling through the BI layer and the subsequent downsampling through the MP_{K_i} layer are removed. The pink dotted curve is obtained when both the pre-processing layers, i.e., the ROF and BI layers are removed. Other parameters remain unchanged in all these scenarios. It can be inferred from Fig. 4 that the both of the pre-processing layers

are essential to obtain a desired level of denoising performance along with the traditional layers of CNN. In addition, the black dashed curve presents the learning curve for the same task, but this time the network is trained by initializing the weights and biases with known data that are obtained from training the network with noise parameters $\sigma = 10$ and $p = 0.15$. It is seen from Fig. 4 that the denoising performance reaches a satisfactory level very quickly in about 10 epochs in the case of transfer learning, thus reducing the training time significantly.

4.4. Learned filters

As per the assigned dimensions of the layers of the CNN model, there are 100, 200, 100 and 1 filters that are obtained for the filter sets W_1, W_2, W_3 and W_4 , respectively, when $\sigma < 20$. Fig. 5 shows the entire set of 100 filters of W_1 learned from the training dataset for SPIN parameters $\sigma = 10$ and $p = 0.30$ and used in the first convolution layer. The weights of any of these filters are combinations of positive and negative real values as shown by the grayscale square-shaped dots (see Fig. 5). These values of weights reveal that each of the filters are bandpass in nature. Similar type of filters also result for W_2, W_3, W_4 , but are not shown in view of space limitations. These filters produce certain bandpass versions of the noisy image in the output channel of the convolution layers. Fig. 6 shows the input noisy image, a few output images resulting from the bandpass filters of the first, second, and third convolution layers and the denoised output image resulting from the final or fourth convolution layer. It is to be pointed out that in the first three convolution layers, the number of outputs are 100, 200, and 100, respectively. But, due to space limitations, we show only 10 typical output images of each of the two layers. It is seen from Fig. 6 that the image features such as the textures and edges are extracted selectively in the convolution layers. The images marked as 1, 2, and 3 in the red color are the low resolution versions of the original image, and such an image is propagated through the network independent of the noise settings. Finally, the output of the fourth convolution layer of the network provides the estimate of the noise-free image.

4.5. Methods used for comparison

The results of the proposed CNN-based method are compared with that of three well established methods when employed to remove mixed-noise. A brief description of the comparing methods are given below:

- Cai's Method [32]: In this method, the impulse noise is detected first as outliers and removed by using AMF. Then, the variational approach-based denoising is employed assuming that the pixels restored in the first phase are essentially free of outliers. The method can also be employed as an ROF for reducing mixed-noise.
- ROF + BM3D [10]: In this method, ROF is first used to remove impulse noise. Then, the shape-adaptive self-similar patches from images are grouped and denoised using the 3D collaborative filtering. In tradition, the median filter is employed as an ROF for performance comparison with this method (see, for example, [36]). In our experiments, the Cai's method is chosen as ROF for mixed AWGN + SPIN. On the other hand, for mixed AWGN + SPIN + RVIN, the choice of ROF is AMF followed by ACWMF to remove SPIN and RVIN sequentially.
- WESNR [36]: The method encodes the noisy image patches over a set of principle component analysis-based dictionaries learned offline and weights the coding residuals to suppress the noise. The weighted encoding is performed by integrating the image sparsity and nonlocal self-similarity priors in a variational framework.

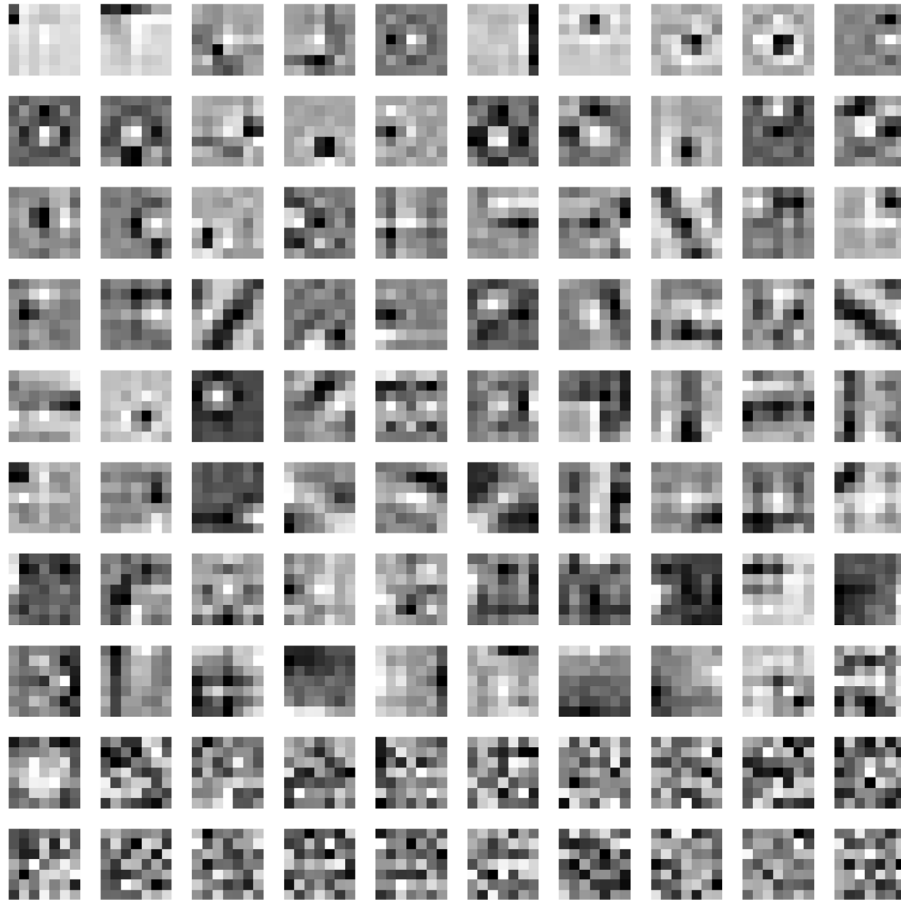


Fig. 5. Grid of the filters in the filter set W_1 learned from the training dataset for SPIN parameters $\sigma = 10$ and $p = 0.30$. The kernel size of each of the filters is 7×7 . The filters are sorted according to the variance of coefficients.

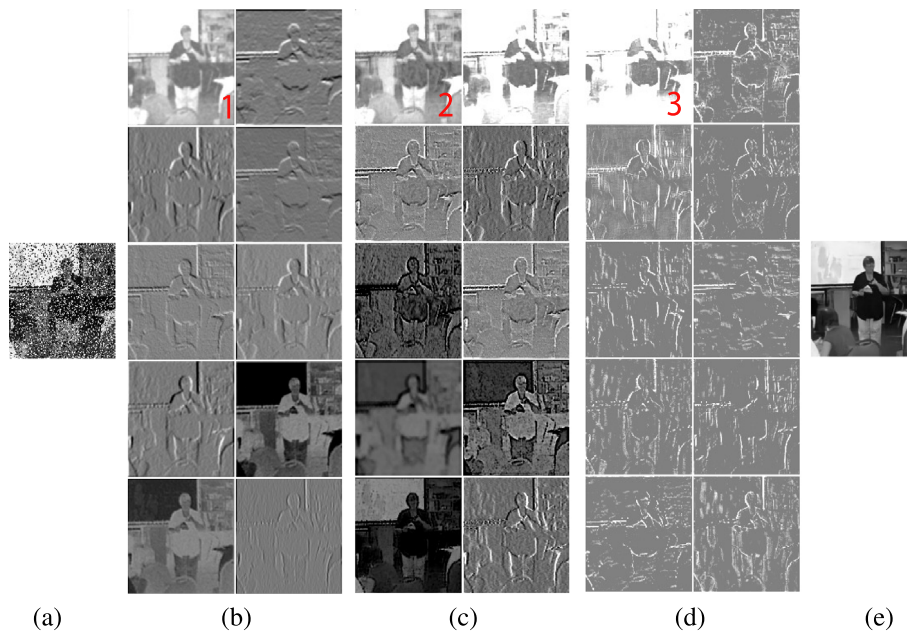


Fig. 6. The input and outputs of different convolution layers for AWGN + SPIN parameters $\sigma = 10$ and $p = 0.30$. (a) Input noisy image. Typical ten outputs of each column are obtained from the convolution layers of (b) first stage, (c) second stage, and (d) third stage. (e) Output of the final convolution layer.

The Cai's, WESNR, and BM3D methods are implemented using the codes distributed by the authors of the respective papers. Default settings prescribed are chosen for the WESNR and BM3D methods. In the

Cai's method, the parameter of out-of-focus kernel is set to zero, since we consider only the noisy version of images. It is the regularization parameter β that determines whether Cai's method would act as an

Table 1

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing mixed AWGN + SPIN from Test Set 1.

Parameters	Cai's [32]	Cai's [32] + BM3D [10]	WESNR [36]	Proposed CNN
$\sigma = 10$	27.18 ± 0.74	30.48 ± 2.11	28.40 ± 2.68	31.28 ± 2.06
$p = 0.15$	0.7618 ± 0.0817	0.9190 ± 0.0281	0.8669 ± 0.0457	0.9256 ± 0.0251
$\sigma = 10$	26.48 ± 1.17	28.76 ± 2.23	27.33 ± 2.89	29.81 ± 2.22
$p = 0.30$	0.7572 ± 0.0701	0.8938 ± 0.0350	0.8500 ± 0.0482	0.9069 ± 0.0316
$\sigma = 15$	24.41 ± 0.41	29.05 ± 2.03	26.58 ± 2.63	29.46 ± 1.97
$p = 0.15$	0.6460 ± 0.1049	0.8831 ± 0.0409	0.8047 ± 0.0751	0.8823 ± 0.0489
$\sigma = 15$	24.24 ± 0.74	27.74 ± 2.13	25.74 ± 2.86	28.40 ± 2.07
$p = 0.30$	0.6464 ± 0.0948	0.8561 ± 0.0469	0.7904 ± 0.0762	0.8694 ± 0.0416
$\sigma = 20$	22.21 ± 0.26	27.85 ± 2.00	25.86 ± 2.59	28.00 ± 1.90
$p = 0.15$	0.5536 ± 0.1127	0.8486 ± 0.0504	0.7850 ± 0.0739	0.8517 ± 0.0457
$\sigma = 20$	22.32 ± 0.48	26.80 ± 2.05	24.99 ± 2.85	27.11 ± 1.97
$p = 0.30$	0.5574 ± 0.01049	0.8208 ± 0.0553	0.7682 ± 0.0744	0.8297 ± 0.0553
$\sigma = 25$	20.40 ± 0.18	26.90 ± 1.98	24.87 ± 2.68	26.93 ± 1.81
$p = 0.15$	0.4798 ± 0.1131	0.8173 ± 0.0580	0.7524 ± 0.0826	0.8151 ± 0.0566
$\sigma = 25$	20.68 ± 0.35	25.99 ± 2.01	23.92 ± 2.97	26.17 ± 1.87
$p = 0.30$	0.4854 ± 0.1071	0.7880 ± 0.0612	0.7344 ± 0.0827	0.7942 ± 0.0562

Table 2

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing mixed AWGN + SPIN from Test Set 2.

Parameters	Cai's [32]	Cai's [32] + BM3D [10]	WESNR [36]	Proposed CNN
$\sigma = 10$	27.20 ± 0.84	30.47 ± 2.13	27.80 ± 2.82	31.29 ± 2.08
$p = 0.15$	0.7628 ± 0.0819	0.9189 ± 0.0288	0.8440 ± 0.0693	0.9253 ± 0.0260
$\sigma = 10$	26.45 ± 1.20	28.74 ± 2.26	26.83 ± 2.93	29.81 ± 2.23
$p = 0.30$	0.7564 ± 0.0721	0.8936 ± 0.0335	0.8283 ± 0.0687	0.9066 ± 0.0320
$\sigma = 15$	24.43 ± 0.59	29.06 ± 2.06	26.57 ± 2.64	29.52 ± 1.98
$p = 0.15$	0.6469 ± 0.1062	0.8831 ± 0.0413	0.8045 ± 0.0751	0.8866 ± 0.0402
$\sigma = 15$	24.24 ± 0.81	27.73 ± 2.12	25.74 ± 3.84	28.41 ± 2.06
$p = 0.30$	0.6464 ± 0.0971	0.8560 ± 0.0476	0.7903 ± 0.0761	0.8688 ± 0.0424
$\sigma = 20$	22.22 ± 0.52	27.86 ± 2.03	25.75 ± 2.64	28.14 ± 1.93
$p = 0.15$	0.5543 ± 0.1141	0.8488 ± 0.0510	0.7790 ± 0.0795	0.8518 ± 0.0474
$\sigma = 20$	22.33 ± 0.65	26.80 ± 2.06	24.89 ± 2.89	27.15 ± 1.99
$p = 0.30$	0.5577 ± 0.1062	0.8207 ± 0.0564	0.7632 ± 0.0809	0.8291 ± 0.0566
$\sigma = 25$	20.40 ± 0.51	26.88 ± 2.00	24.85 ± 2.67	26.95 ± 1.85
$p = 0.15$	0.4794 ± 0.1144	0.8169 ± 0.0589	0.7520 ± 0.0829	0.8131 ± 0.0602
$\sigma = 25$	20.68 ± 0.60	25.98 ± 2.05	23.90 ± 3.03	26.13 ± 1.90
$p = 0.30$	0.4855 ± 0.1071	0.7874 ± 0.0621	0.7329 ± 0.0835	0.7922 ± 0.0612

Table 3

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing mixed AWGN + SPIN from Test Set 3.

Parameters	Cai's [32]	Cai's [32] + BM3D [10]	WESNR [36]	Proposed CNN
$\sigma = 10$	27.21 ± 0.83	31.02 ± 2.96	27.84 ± 4.26	31.61 ± 2.75
$p = 0.15$	0.6794 ± 0.0856	0.8650 ± 0.0491	0.8414 ± 0.0536	0.9067 ± 0.0348
$\sigma = 10$	26.53 ± 1.35	29.21 ± 3.13	27.29 ± 4.79	29.98 ± 2.55
$p = 0.30$	0.6787 ± 0.0740	0.8788 ± 0.0418	0.8233 ± 0.0560	0.9116 ± 0.0329
$\sigma = 15$	24.34 ± 0.44	29.11 ± 2.32	25.73 ± 3.64	29.48 ± 2.25
$p = 0.15$	0.5703 ± 0.0655	0.8631 ± 0.0474	0.7744 ± 0.0852	0.8642 ± 0.0461
$\sigma = 15$	24.16 ± 0.76	27.73 ± 2.35	24.98 ± 3.85	28.38 ± 2.26
$p = 0.30$	0.5731 ± 0.1236	0.8331 ± 0.0557	0.7619 ± 0.0867	0.8489 ± 0.0468
$\sigma = 20$	22.20 ± 0.47	28.59 ± 2.79	25.75 ± 4.09	28.60 ± 2.42
$p = 0.15$	0.4482 ± 0.1011	0.8322 ± 0.0569	0.7622 ± 0.0852	0.8234 ± 0.0540
$\sigma = 20$	22.35 ± 0.53	27.47 ± 3.83	25.02 ± 4.27	27.66 ± 2.50
$p = 0.30$	0.4529 ± 0.0945	0.8017 ± 0.0617	0.7439 ± 0.0894	0.8072 ± 0.0539
$\sigma = 25$	20.38 ± 0.18	27.11 ± 2.12	24.37 ± 3.58	27.18 ± 2.00
$p = 0.15$	0.4009 ± 0.0658	0.7961 ± 0.0652	0.7265 ± 0.0941	0.7953 ± 0.0597
$\sigma = 25$	20.66 ± 0.33	26.16 ± 2.10	23.57 ± 3.65	27.04 ± 2.01
$p = 0.30$	0.4056 ± 0.0605	0.7641 ± 0.0673	0.7118 ± 0.0952	0.7845 ± 0.0610

Table 4

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing mixed AWGN + SPIN + RVIN from Test Sets 1, 2, and 3.

Parameters	Test Sets	AMF + ACWMF + BM3D [10]	WESNR [36]	Proposed CNN
$\sigma = 10$ $p = 0.20$ $r = 0.05$	Test Set 1	26.52 ± 2.68	26.70 ± 2.73	26.96 ± 2.67
		0.8407 ± 0.0542	0.8281 ± 0.0574	0.8565 ± 0.0468
	Test Set 2	26.73 ± 2.73	26.49 ± 2.79	27.18 ± 2.75
		0.8493 ± 0.0498	0.8156 ± 0.0720	0.8605 ± 0.0469
	Test Set 3	26.60 ± 3.09	26.45 ± 3.18	26.93 ± 3.07
		0.8254 ± 0.0631	0.8086 ± 0.0630	0.8365 ± 0.0568
$\sigma = 20$ $p = 0.15$ $r = 0.10$	Test Set 1	24.87 ± 2.28	24.84 ± 2.39	25.13 ± 2.27
		0.7523 ± 0.0694	0.7448 ± 0.0821	0.7677 ± 0.0690
	Test Set 2	25.15 ± 2.32	24.98 ± 2.46	25.33 ± 2.31
		0.7627 ± 0.0680	0.7495 ± 0.0850	0.7727 ± 0.0695
	Test Set 3	24.82 ± 2.50	24.79 ± 2.70	25.06 ± 2.48
		0.7235 ± 0.0802	0.7229 ± 0.0928	0.7377 ± 0.0822

independent method for removal of mixed-noise or as an ROF. In the experiments, β of this method is chosen in such a way that the overall denoising performance is the best. The codes of the proposed CNN-based denoising method and the test images for which the results are reported are shared in the web-link given in [52].

4.6. Results

In the experiments, the denoising performance of the methods are compared in three ways. First, the PSNR and the SSIM indices [49] are used to assess the denoising performance quantitatively. Next, the qualitative evaluation is performed by observing the outputs of the methods, i.e., the denoised images. Finally, the methods are compared in terms of the computational load to denoise an image.

4.6.1. Quantitative evaluation

Tables 1–3 show the average denoising performance for different settings of parameters of mixed AWGN + SPIN in Test Sets 1, 2 and 3, respectively, in terms of the metrics PSNR and SSIM for the methods being compared. It is seen from Table 1 that the average values of PSNR and SSIM considering the 1000 images in Test Set 1 are the highest among the comparing methods in most of the cases (14 out of 16 instances). For example, the improvement of the CNN-based method over the closest one, Cai's + BM3D, is approximately 2.6% in terms of average values of PSNR and it is 0.72% in terms of that of SSIM for $\sigma = 10$ and $p = 0.15$. As the strength of noise is increased, the proposed method consistently performs better than the methods compared in terms of mean of PSNR. It is also observed from the table that the standard deviation of the metric PSNR obtained by the proposed method is always the lowest and that of the metric SSIM is the lowest for 7

Table 6

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing AWGN from Test Set 1.

Parameter	BM3D [10]	WESNR [36]	Proposed CNN
$\sigma = 10$	33.93 ± 1.81	30.63 ± 2.57	33.50 ± 1.86
	0.9388 ± 0.0240	0.9050 ± 0.0267	0.9438 ± 0.0222
$\sigma = 15$	30.53 ± 1.88	28.00 ± 2.80	31.14 ± 1.90
	0.9034 ± 0.0361	0.8520 ± 0.0567	0.9110 ± 0.0336
$\sigma = 20$	28.93 ± 1.92	26.80 ± 2.90	29.51 ± 1.93
	0.8703 ± 0.0462	0.8236 ± 0.0551	0.8804 ± 0.0443
$\sigma = 25$	27.74 ± 1.94	25.04 ± 3.30	28.31 ± 1.93
	0.8395 ± 0.0543	0.7836 ± 0.0689	0.8509 ± 0.0533

out of 8 instances in comparison to the other methods, thus providing the most robust denoising performance. For example, when the noise parameters are $\sigma = 10$ and $p = 0.15$, the gains in robustness over the closest competing method Cai's + BM3D are 2.4% for PSNR and 10.7% for SSIM. Similarly, the proposed CNN-based method provides the overall best denoising performance in terms of the average of PSNR and SSIM for removal of AWGN + SPIN from 1000 images of Test Set 2 and 15 images of Test Set 3, as observed from Tables 2 and 3, respectively. In particular, the improvements of the proposed method over Cai's + BM3D in terms of the performance metrics are observed for 30 and 31 instances out of 32 for the Test Sets 2 and 3, respectively. In summary, it can be inferred from Tables 1 to 3 that, in general, the closest competitor of the proposed CNN-based method for removal of mixed AWGN + SPIN from the Test Sets is the method Cai's + BM3D.

Table 4 presents the comparative performance in terms of the metrics PSNR and SSIM for the challenging case of reducing the mixed AWGN + SPIN + RVIN from Test Sets 1, 2 and 3. It can be observed from the table that the overall performance in terms of the mean and standard deviation of the metrics is superior for the proposed CNN-based method. In the case when the strength of noise is relatively low, i.e., $\sigma = 10$, $p = 0.20$ and $r = 0.05$, the improvement in average of PSNR is 0.97% and that of SSIM is 1.88% over the competitive method WESNR and AMF + ACWMF + BM3D in Test Set 1. The improvement in robustness in this case is 0.35% in PSNR and 13.7% in SSIM. For a higher level noise with $\sigma = 20$, $p = 0.15$ and $r = 0.15$, the methods AMF + ACWMF + BM3D and WESNR show a performance that is competitive with the proposed method in terms of the metrics of average of PSNR and SSIM, respectively. For example, the improvement in average of PSNR is 1% and that in SSIM is 2% over AMF + ACWMF + BM3D in Test Set 1. In this scenario, the improvements in robustness are also significant with 1.3% in PSNR and 0.57% in SSIM. Similar improvements (14 out of 16 instances) in denoising performance for reducing the AWGN + SPIN + RVIN are also observed for Test Sets 2 and 3.

Table 5

Denoising performance in terms of mean and standard deviation of PSNR (in dB) and SSIM for reducing mixed AWGN + SPIN and AWGN + SPIN + RVIN from commonly-referred five test images.

Test Images	AWGN + SPIN ($\sigma = 20$ and $p = 0.30$)			AWGN + SPIN + RVIN ($\sigma = 20$, $p = 0.15$ and $r = 0.10$)		
	Cai's [32] + BM3D [10]	WESNR [36]	Proposed CNN	AMF + ACWMF + BM3D [10]	WESNR [36]	Proposed CNN
Peppers	29.57 ± 1.58	28.52 ± 1.73	29.99 ± 1.38	27.98 ± 1.69	28.25 ± 1.81	28.37 ± 1.67
	0.8601 ± 0.0295	0.8462 ± 0.0232	0.8634 ± 0.0317	0.8290 ± 0.0266	0.8404 ± 0.0203	0.8425 ± 0.0266
Goldhill	28.64 ± 0.81	27.37 ± 0.93	28.81 ± 0.69	27.50 ± 0.79	27.30 ± 0.94	27.61 ± 0.74
	0.7810 ± 0.0117	0.7088 ± 0.0119	0.7879 ± 0.0181	0.7360 ± 0.0104	0.7110 ± 0.0127	0.7377 ± 0.0124
Boat	28.92 ± 1.12	26.45 ± 1.16	28.36 ± 0.90	26.35 ± 1.06	26.24 ± 0.79	26.57 ± 1.01
	0.7894 ± 0.0121	0.7197 ± 0.0241	0.8024 ± 0.0131	0.7355 ± 0.0143	0.7360 ± 0.0103	0.7432 ± 0.0126
Man	28.08 ± 1.10	26.97 ± 1.10	28.43 ± 0.94	26.89 ± 1.05	26.92 ± 1.06	27.14 ± 0.97
	0.7958 ± 0.0027	0.7451 ± 0.0053	0.8075 ± 0.0085	0.7534 ± 0.0038	0.7450 ± 0.0042	0.7610 ± 0.0037
Baboon	25.16 ± 0.69	23.71 ± 1.08	25.61 ± 0.63	23.71 ± 1.11	23.65 ± 1.05	23.87 ± 1.08
	0.6862 ± 0.0158	0.5569 ± 0.0217	0.7223 ± 0.0141	0.5970 ± 0.0129	0.5690 ± 0.0209	0.6027 ± 0.0104
Average	27.88 ± 1.06	26.61 ± 1.2	28.24 ± 0.91	26.48 ± 1.16	26.47 ± 1.13	26.71 ± 1.07
	0.7825 ± 0.0144	0.7154 ± 0.0172	0.7968 ± 0.0171	0.7302 ± 0.0134	0.7203 ± 0.0136	0.7374 ± 0.0131

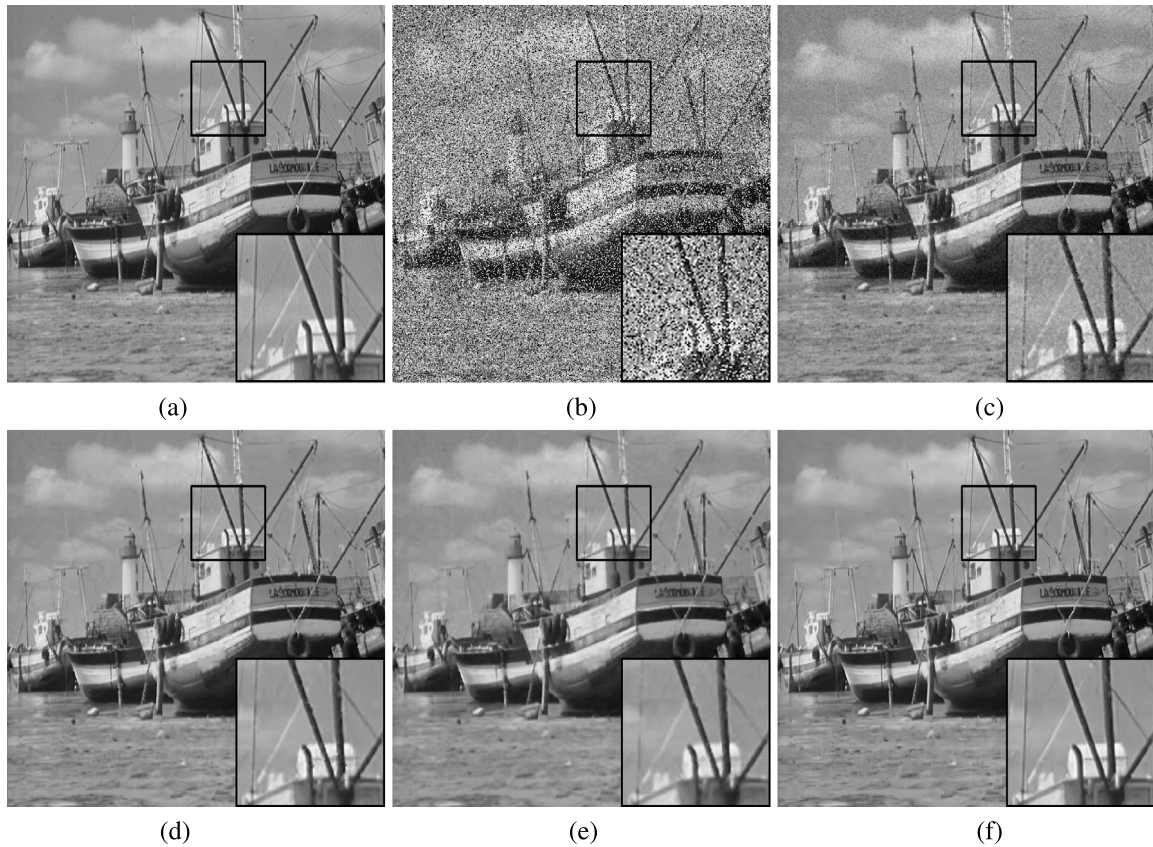


Fig. 7. Visual comparison of the denoising performance of the methods for the test image *Boat*. (a) The ground truth of noise-free image. (b) The image is corrupted by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$ (PSNR: 10.65 dB, SSIM: 0.0731). (c) The image is obtained by using Cai's method (PSNR: 27.70 dB, SSIM: 0.6929). The estimated noise-free images are obtained by using the methods (d) Cai's + BM3D [10] (PSNR: 31.21 dB, SSIM: 0.8529), (e) WESNR [36] (PSNR: 29.56 dB, SSIM: 0.8062), and (f) proposed CNN (PSNR: 31.92 dB, SSIM: 0.8596).

In order to evaluate the denoising performance of the proposed CNN-based method on commonly-referred test images, we choose five classic grayscale images based on the details in the scene. The images are *Peppers*, *Goldhill*, *Boat*, *Man* and *Baboon*, which have increasing details in the same order. In accordance with the spirit of Test Set 3, the images are considered in three different sizes, namely, 128×128 , 256×256 and 512×512 . First, the images are corrupted with mixed AWGN + SPIN using the noise parameters $\sigma = 20$ and $p = 0.30$. Then, they are denoised using the methods Cai's + BM3D, WESNR and proposed CNN. This procedure is repeated 5 times and the average results for each of the images are presented in Table 5. It is observed from the table that the improvements in terms of PSNR and SSIM resulting from the proposed method over Cai's + BM3D for the relatively smooth image *Peppers* are 1.4% and 0.38%, respectively. The increasing improvement of denoising performance for mixed AWGN + SPIN is observed for the proposed method with the increasing details of images. For example, the improvements in the performance metrics PSNR and SSIM are found to be 1.7% and 5.2%, respectively, for the image *Baboon* that has significant details. It is also evident from the table that standard deviation of the performance metrics is consistently low for the proposed method, thus making it robust denoising method. A similar experiment is conducted for reducing the mixed AWGN + SPIN + RVIN with parameters $\sigma = 20$, $p = 0.15$ and $r = 0.10$, and compared against the methods AMF + ACWMF + BM3D and WESNR. It is also observed that the denoising performance of the proposed method increases with increasing details of the scene for reducing the mixed AWGN + SPIN + RVIN from images.

To verify the effectiveness of the proposed CNN model in reducing AWGN from images, experiments are conducted by excluding the rank order filtering part of the proposed method and that of the WESNR [36].

The performance of the proposed CNN model is also compared with the BM3D method [10], which is well-known for removal of AWGN from images. Table 6 presents the denoising performance in terms of PSNR and SSIM for reducing AWGN from Test Set 1. It can be observed from this table that the proposed CNN-based method performs consistently better than the WESNR method. In terms of the mean of the metrics, the proposed method is unbeatable with respect to the BM3D method, but it shows competitive performance in terms of robustness at the low level of noise. Similar results are found for the other test data, but are not included in the paper to avoid presenting repetitive results.

4.6.2. Qualitative evaluation

Examples of commonly-referred test images as well as that of images with close-up and long-shot views are considered for the evaluation of the qualitative performance. Fig. 7 shows the ground truth of noise-free version of the *Boat* image, its noisy version corrupted by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$, and the denoised images obtained by using the Cai's, Cai's + BM3D, WESNR and proposed CNN-based methods. In this image there are regions with textures such as the 'sands', regions with sharp edges such as the 'masts' of the boats, smooth region such as the 'sky', and text region such as the 'name' of the boat. It can be observed from the denoised images that the proposed CNN reconstructs most of the details such as that of the mast and roof of the engine-room (see zoomed-in region of the image).

Fig. 8 shows ground truth of noise-free version of a high quality image having a close-up view taken from the Test Set 3, its noisy version corrupted by mixed AWGN + SPIN with parameters $\sigma = 25$ and $p = 0.15$, and the denoised images obtained by using the comparing methods. This image shows a picture of a 'butterfly' in a rectangular 'frame' along with the 'text' of copyright in the bottom. It is seen from Fig. 8

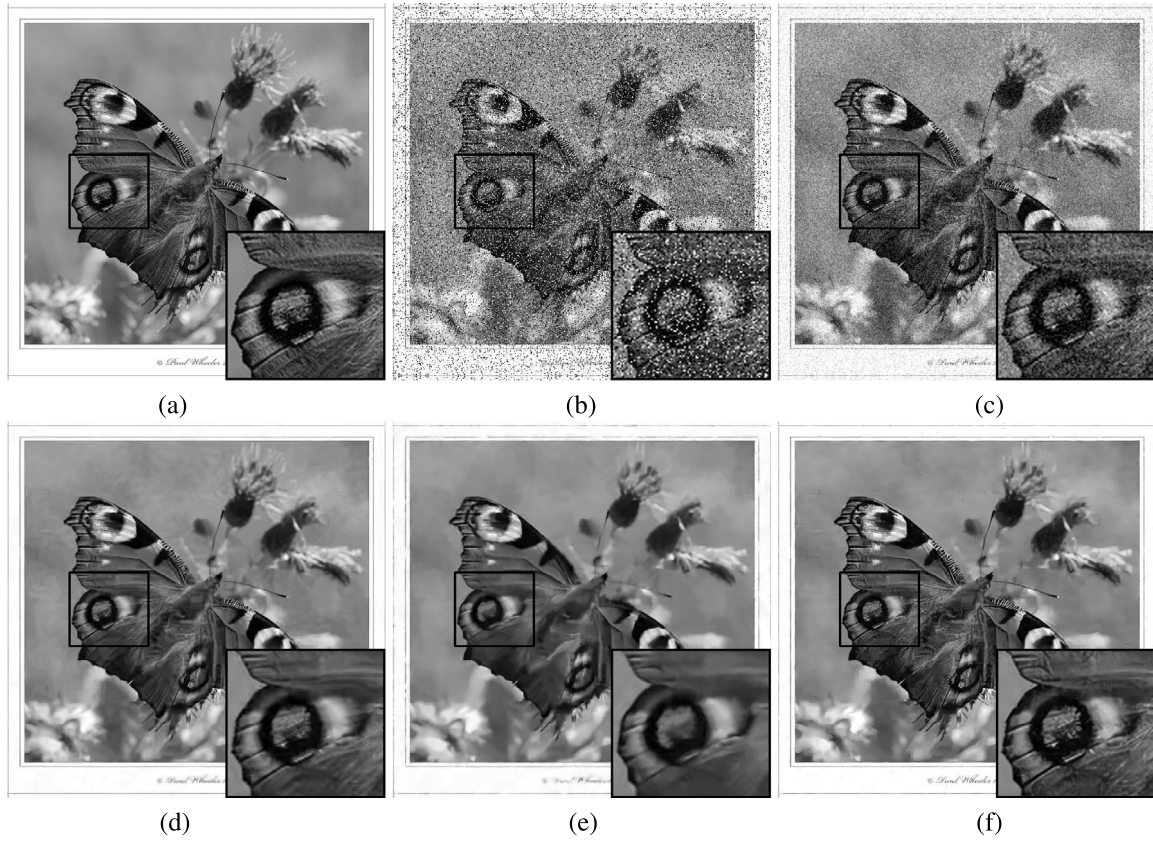


Fig. 8. Visual comparison of the denoising performance of the methods on a high quality image having a close-up view. (a) The ground truth of noise-free image. (b) The image is corrupted by mixed AWGN + SPIN with parameters $\sigma = 25$ and $p = 0.15$ (PSNR: 12.40 dB, SSIM: 0.1419). (c) The image is obtained by using Cai's method (PSNR: 20.40 dB, SSIM: 0.3587). The estimated noise-free images are obtained by using the methods (d) Cai's + BM3D [10] (PSNR: 28.71 dB, SSIM: 0.8838), (e) WESNR [36] (PSNR: 25.77 dB, SSIM: 0.8458), and (f) proposed CNN (PSNR: 28.81 dB, SSIM: 0.8906).

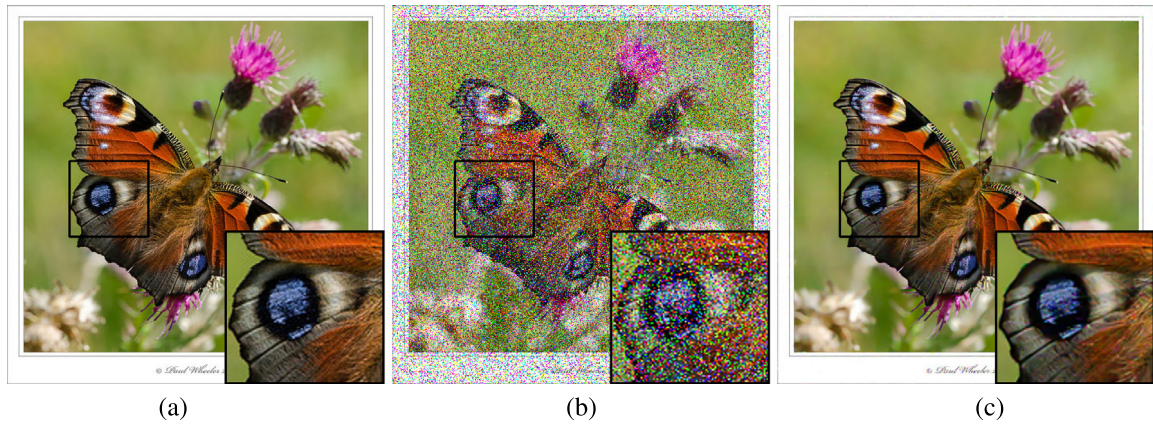


Fig. 9. Visual comparison of denoising performance of the proposed method for the color version of the high quality image. (a) The ground truth of noise-free image. (b) The image is corrupted by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$ (PSNR: 5.47 dB, SSIM: 0.2466). (c) The estimated noise-free image is obtained by using the proposed CNN (PSNR: 24.79 dB, SSIM: 0.9296).

that the image restored by using the competitive Cai's + BM3D method shows a good quality edge preservation, when similar edges are present throughout the image such as the frame. For edges and textures that are not repeated, such as the text of copyright or the details of the wings of butterfly, the proposed CNN method clearly provides superior denoising performance. In order to test the performance of the proposed denoising method on the color version of the image, the trained CNN model is used to reduce noise from the red, green and blue channels of the image. Fig. 9 shows the visual output of the color image, its noisy version corrupted by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$, and the

denoised image obtained by the proposed CNN model. It is seen from this figure that the proposed denoising method can successfully reduce mixed-noise from the color version of the image with a performance very similar to that obtained by applying it to the grayscale version.

Fig. 10 shows a typical image having a long shot view with its noise-free and noisy versions corrupted by mixed AWGN + SPIN + RVIN having parameters $\sigma = 10$, $p = 0.20$ and $r = 0.05$, and the corresponding denoised images obtained by using the AMF + ACWMF + BM3D, WESNR, and proposed CNN methods. Most of the parts of this image have heavy details such as those in 'trees', 'pedestrians', and 'horse' with

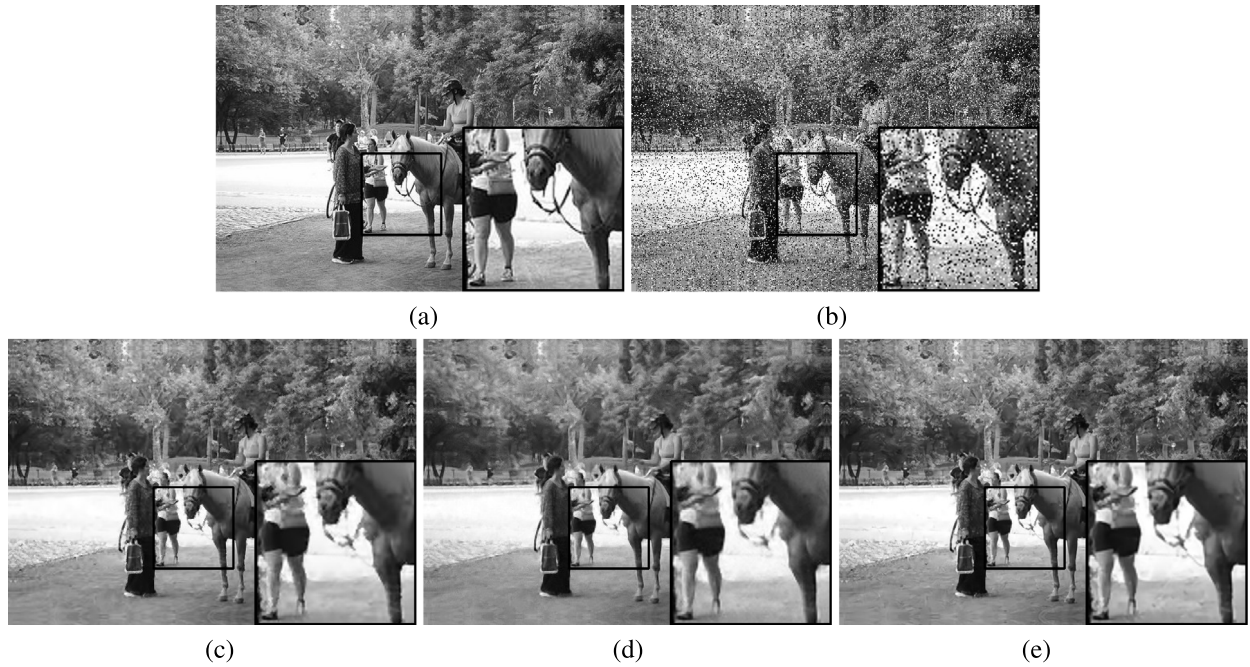


Fig. 10. Visual comparison of the denoising performance of the methods for mixed AWGN + SPIN + RVIN denoising for an image having long shot view. (a) The ground truth of noise-free image. (b) The image is corrupted by mixed AWGN + SPIN + RVIN with parameters $\sigma = 10$, $p = 0.20$ and $r = 0.05$ (PSNR: 12.75 dB, SSIM: 0.2316). The estimated noise-free images are obtained by using the methods (c) AMF + ACWMF + BM3D [10] (PSNR: 22.64 dB, SSIM: 0.6893), (d) WESNR [36] (PSNR: 22.44 dB, SSIM: 0.6592), and (e) proposed CNN (PSNR: 22.83 dB, SSIM: 0.7088).

rider. It can be observed from this figure that the image recovered by using AMF + ACWMF + BM3D has jagged edges (see, for example, leg of ‘horse’) and that by WESNR is considerably blurred (see, for example, reins of ‘horse’). It is seen from the figure that the image recovered by using the proposed CNN-based method resembles original image the most as compared to that recovered by using the method WESNR or AMF + ACWMF + BM3D (see zoomed-in region of the image).

4.6.3. Evaluation of computation time

Experiments are carried out using a computer equipped with an Intel i7 4770K 3.5 GHz processor, 16GB of memory and an NVIDIA Titan Xp GPU. The operating system of the computational environment is Ubuntu 16.04. The three methods being compared, namely, Cai’s, WESNR, and ROF + BM3D, are executed in the MATLAB 2016b Linux edition, whereas the proposed CNN is trained in the Tensorflow platform. Fig. 11 shows a comparative bar-chart representing the mean execution time in seconds required by the four methods for denoising an image of size 128×128 contaminated by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$. It can be seen from this figure that the proposed CNN-based method performs the denoising task in less than a fraction of a second. The proposed method is 2.6 times faster than the closest competitor Cai’s + BM3D and 36 times faster than the WESNR method. It is to be noted that the AMF + ACWMF requires only few milliseconds over Cai’s method and thus is not included in the comparison. Such a small computation time with superior denoising performance put the proposed CNN-based method to be in a favorable position in comparison to the existing methods.

5. Conclusions

In this paper, a CNN-based deep learning approach has been proposed for reduction of mixed Gaussian-impulse noise from images. Restoration of images contaminated by the AWGN and IN is still a challenge even for the handcrafted algorithms due to their contrasting natures. In this context, a learning-based approach has been adopted such that the necessary restoration process of mixed Gaussian-impulse noise can be developed through extensive training with large number of

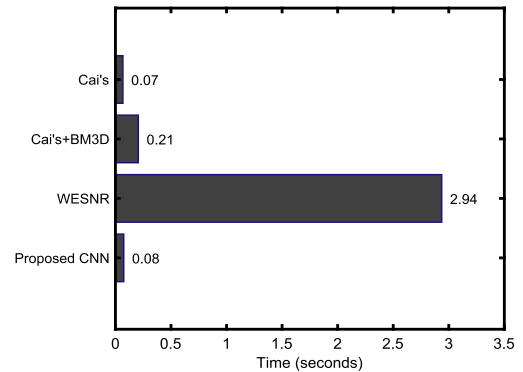


Fig. 11. Comparison of execution time of the four experimental methods required to denoise an image of size 128×128 contaminated by mixed AWGN + SPIN with parameters $\sigma = 10$ and $p = 0.30$.

image variabilities. Among the different approaches of machine learning algorithms, CNN is particularly suitable for image processing and has shown vast representation capabilities. The architecture of CNN is such that the 2D structure of local neighboring regions of images remains unaltered as opposed to the vectorization of image or patch matrices employed in the sparse coding-based methods. Further, the possibility of utilization of an effective number of self-similar patches gathered from huge size training set through a gradient descent learning in the CNN-based algorithm is much higher as compared to that in the case of traditional non-local means or patch-based methods, which collect a sub-optimal set of patches from the test images. In the proposed method, the image corrupted by mixed Gaussian-impulse noise has been preprocessed by rank order filtering and upsampled by means of BI. The filtered and interpolated image is fed to a 4-stage CNN architecture, wherein each stage consists of a suitable set of layers including the convolution, ReLU and max-pooling layers. The dimensions of the four sets of convolution filters and the corresponding bias terms are chosen

as per the size of the input and the output of the interconnected layers and that of the local neighboring mask used in each of the layers. The weights of the filters and biases in the four stages of CNN have been learned from a sufficiently large dataset with the adoption of suitable data augmentation techniques. Experimentation on the validation set of images has revealed that the use of BI layer, which acts as a low-level smoother in the preprocessing stage, significantly improves the overall performance of the CNN-based denoising scheme. A faster training is achieved by adopting the mechanism of ‘transfer learning’, wherein the known weights and biases of a network for a particular setting of noise parameters are used to initialize the same for training the network in a new setting of noise parameters. The proposed method has been extensively tested against well established methods, namely, Cai’s method, ROF+BM3D, and WESNR using three different test datasets and commonly-used individual test images considering different practical scenarios. In particular, not only different imaging environments of the training and testing sets have been considered in the experiments, but also denoising experiments on different settings of mixed-noise scenarios including AWGN+SPIN and AWGN+SPIN+RVIN have been carried out. The strengths of the Gaussian and impulse noise and their mixing proportion have been varied widely to evaluate the effectiveness of the proposed CNN-based model for reducing the mixed-noise. In addition, the effect due to deviations of noise parameters and that of the sizes of images during training and testing phases have been examined. It has been shown that the average values of the commonly-used performance metrics, namely, the PSNR and SSIM for reducing mixed AWGN+SPIN or AWGN+SPIN+RVIN are the highest in almost all cases for the proposed CNN-based method among all the methods compared. The overall robustness of the denoising performance of the proposed method has been found to be significantly superior in terms of these metrics. The visualization of denoised images has also revealed that the proposed method retains the image structure and details maximally as compared to the other methods both in reducing mixed AWGN+SPIN and AWGN+SPIN+RVIN. In addition, it has been observed that the proposed method requires a very insignificant computational time for denoising an image. In conclusion, the light weight structure of the proposed CNN model can play a significant role in many applications, where the low-complexity and robust denoising performance are primary concerns. In future, similar CNN model can be employed in various closely related regression type problems such as image inpainting and watermark removal.

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